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STUDIES ON SUCCINATE DEHYDROGENASE

I. SPECTRAL PROPERTIES OF THE PURIFIED ENZYME AND FORMATION OF ENZYME-COMPETITIVE INHIBITOR COMPLEXES

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SUMMARY

1. Succinate dehydrogenase (succinate:(acceptor) oxidoreductase, EC 1.3.99.1) reacts immediately with competitive inhibitors to form spectrally detectable enzyme-inhibitor complexes, which are converted to succinate-reduced enzyme on the subsequent addition of succinate.

2. The enzyme-inhibitor complexes formed with malonate, fumarate, maleate and methylene succinate differ from that with oxaloacetate in spectral characteristics, the latter showing a wide, diffuse band from 500 to 750 m μ .

3. Pyrophosphate, a strong competitive inhibitor of succinate dehydrogenase, but which is not in any obvious way structurally related to succinate, has no effect on the spectrum.

4. Dissociation constants of the enzyme-inhibitor complexes determined by spectral titration are in good agreement with kinetically determined inhibitor constants.

5. The changes in spectrum associated with the formation of the enzyme-inhibitor complexes are explainable in terms of changes in polarity near the flavin prosthetic group and by the formation of charge-transfer complexes between the electron-donating inhibitor and the electron-accepting enzyme.

6. The apparent incomplete reoxidation by fumarate of dithionite-reduced enzyme, reported by previous workers, can be explained by the formation of the complex between oxidized enzyme and fumarate.

INTRODUCTION

It is well-established that succinate dehydrogenase (succinate:(acceptor) oxidoreductase, EC 1.3.99.1) is a flavoprotein^{1,2}, the prosthetic group of which is FAD bound to the protein through the isoalloxazine ring^{3,4}. In addition, it contains four iron atoms per molecule of 150 000 (ref. 2) or 200 000 (ref. 1).

Although it is to be expected that the flavin prosthetic group plays a role in the enzymic reaction, this has proved difficult to demonstrate convincingly. Thus, both SINGER *et al.*¹ and WANG *et al.*² found that the addition of the substrate to the enzyme caused only a slight decrease (15–18%) of absorption in the region of the spectrum where flavin absorbs. Although this is also the case with such flavoproteins as lipo-

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amide dehydrogenase (EC 1.6.4.3) where reduction by substrate proceeds only to the semiquinone level⁵, increased absorption in the region of $530\text{ m}\mu$, characteristic of semiquinone formation, is not found on the addition of succinate to succinate dehydrogenase. MASSEY AND SINGER⁶ showed that addition of $\text{Na}_2\text{S}_2\text{O}_4$ to the enzyme caused a much larger decline of the absorption at $450\text{ m}\mu$ than obtained with succinate and that the absorbancy increased again on the addition of fumarate but not to its original level. Because of the small spectral effect of succinate and the incomplete oxidation by fumarate, KEILIN AND KING⁷ found it difficult to accept that the spectral changes associated with the addition of succinate to the oxidized enzyme or of fumarate to the $\text{Na}_2\text{S}_2\text{O}_4$ -reduced enzyme reflect the oxido-reduction of the flavin in the enzyme.

For these reasons, the spectral effects of fumarate on the enzyme were studied in some detail. It was found that fumarate, as well as a number of other competitive inhibitors of the enzyme, formed spectrophotometrically recognizable complexes with the flavoprotein and that the dissociation constant of the complex is of the same order as the kinetically determined inhibition constants (K_i)

A preliminary report has appeared elsewhere⁸. The experiments have also been discussed in a symposium lecture⁹

MATERIALS AND METHODS

Preparation of succinate dehydrogenase

Since the modification of the method of WANG *et al.*² used in this laboratory for the preparation of succinate dehydrogenase has not been published, it will be given here in detail. The method is similar to that of KEILIN AND KING⁷

Mince from 30 pig hearts (about 3.5 kg) was washed with about 8–10 changes of tap water at 0° , and was then stirred overnight in 25 l 0.15 M phosphate buffer (pH 7.4) at 0° . The mince was further washed twice with tap water at 0° and a KEILIN AND HARTREE¹⁰ heart-muscle preparation made as described by SLATER¹¹. The preparation was taken up in 50 mM borate–50 mM phosphate buffer (pH 8.0) containing 1 mM EDTA, so that the final suspension contained 10 mg protein/ml. This preparation was allowed to stand 12–36 h at 0 – 4° before extraction of the succinate dehydrogenase

The preparation was made anaerobic by repeated evacuation and flushing with N_2 . Succinate (0.8 M) was then added to a final concentration of 40 mM and the evacuation and flushing with N_2 was repeated. The suspension was allowed to stand at 0° for 2–48 h (usually 2 h). All further treatment was carried out at 0° , unless specified otherwise. The pH was brought to 9.0 by the careful addition of 1 N NaOH, and 0.2 volume (calculated on original volume of heart-muscle preparation) butanol at -20° was added slowly with shaking. The mixture was shaken occasionally during 30 min and then centrifuged for 15 min at $1200 \times g$ in polyethylene cups in an International Centrifuge Model PR-2. The middle yellow-orange aqueous layer was carefully withdrawn by an adjustable suction system*. The pH was carefully adjusted to 6.0 with 1 N acetic acid and $\text{Ca}_3(\text{PO}_4)_2$ gel¹² was added to a final concentration of

* This is the crucial step of the procedure. Contamination with butanol-denatured protein leads to an impure preparation.

4 mg/ml. After stirring for 5 min, the suspension was centrifuged for 4 min at $1000 \times g$ in glass tubes. The supernatant was discarded and the gel washed once by stirring in twice-distilled water, de-oxygenated by passage of a stream of N_2 . Elution was carried out by homogenizing the gel with a glass homogenizer under nitrogen in 75 mM phosphate buffer (pH 7.8), containing 1 mM EDTA*. After standing 10 min with occasional shaking, the suspension was centrifuged for 10 min at $23\,000 \times g$ in a Lourdes refrigerated centrifuge No. LRA. Polyethylene cups with caps were used and nitrogen was passed through the suspension before closing the tubes. This high-speed centrifugation removed any butanol-denatured protein which was bound to the gel. The eluate was adjusted to pH 7.2 with 1 N acetic acid and solid ammonium sulfate was slowly added to 60% saturation (at 25°). After 5 min nitrogen was passed through the suspension which was then centrifuged for 10 min at $23\,000 \times g$. The precipitate was taken up in 20 ml of 75 mM phosphate buffer (pH 7.2), containing 1 mM EDTA. A solution saturated with ammonium sulphate, and adjusted to pH 7.2 with ammonia, was added dropwise to 30% saturation (at 25°) or to the first sign of turbidity. After 5 min, the suspension was centrifuged for 5 min at $30\,000 \times g$. The supernatant was brought to 40–42% saturation with the ammonium sulphate solution, nitrogen was passed through the solution and the precipitate was collected by centrifugation in closed cups for 5 min at $30\,000 \times g$, and was dissolved in 0.1 M phosphate buffer (pH 7.6), containing 1 mM EDTA. The amber-coloured solution (25–35 mg protein/ml) was immediately placed in a plastic tube with a self-sealing rubber stopper through which a needle was inserted. The contents of the tube were subjected to 5–6 cycles of evacuation and refilling with N_2 , ending with the latter. The needle was withdrawn and the enzyme solutions were stored at 77°K. Such frozen preparations, if not disturbed, were stable for at least three weeks. When required, the contents of a tube were thawed at 0° and were then centrifuged under N_2 for 5 min at $30\,000 \times g$ at 0°.

The activity of the enzyme, measured by KEILIN AND KING's⁷ modification of WANG *et al.*'s² ferricyanide method, was 3–4 μ moles succinate oxidized/min/mg protein at 25°. The activity by the ferricyanide method is for comparative purposes also expressed in units/mg protein as defined by WANG *et al.*². (The protein was determined by the biuret method¹³.) Measured manometrically with phenazine methosulfate as acceptor, according to the procedure of KEARNEY AND SINGER¹⁴, a typical preparation oxidized 18–20 μ moles succinate/min/mg protein. These activities are of the same order as those given by KING¹⁵ and are higher than those reported by SINGER *et al.*¹.

Determination of inhibitor constants (K_i)

The procedure adopted was essentially that of SLATER AND BONNER¹⁶, using the KEILIN AND HARTREE heart-muscle preparation^{10,11} or the soluble enzyme as source of enzyme and ferricyanide as acceptor. The reaction mixture (*cf.* ref. 7) contained 100 mM potassium phosphate buffer (pH 7.8), 1 mM EDTA, 1 mg/ml bovine serum albumin, 10 mM KCN (absent for the soluble enzyme) and 6 mM $K_3Fe(CN)_6$. Using varying amounts of succinate and a fixed concentration of inhibitor, the initial rates

* All buffer solutions used after solubilization of the enzyme were passed through Dowex A-1 (acid form) and were de-oxygenated by a stream of nitrogen.

of reduction of the $K_3Fe(CN)_6$ were measured at 420 m μ , using a thermostated Zeiss M4Q111 spectrophotometer fitted with a Gilford automatic cuvette changer (Model 210) and Optical Density Converter (Model 200) linked with a Leeds and Northrup Speedomax H recorder. The K_i was calculated from the double reciprocal plot in the presence and absence of inhibitor.

Determination of dissociation constants (K_D)

Dissociation constants were determined spectrophotometrically in 1-cm cells of 1 ml capacity in a Cary Model 14 recording spectrophotometer, thermostated at 25°. Varying concentrations of inhibitor were added to a constant amount of enzyme in 0.1 M phosphate buffer (pH 7.6), containing 1 mM EDTA, and the increase of absorbancy at 510 m μ was measured. The concentration giving half-maximal increase of $A_{510\text{ m}\mu}$ was taken as the dissociation constant (K_D). In the case of malonate and oxaloacetate, which are more firmly bound to the enzyme than other inhibitors, the concentration of free inhibitor was taken to be total concentration of inhibitor added minus concentration of enzyme.

Reagents

Alloxan hydrate, α -oxobutyric acid, L-tartaric acid (Lot 51769 92K), D-tartaric acid (Lot 53817 62K) and meso-tartaric acid (32887 112K) were used as supplied by Fluka; L-aspartic acid, crotonic acid, citric acid, diacetyl, EDTA, $K_3Fe(CN)_6$, glycerol, glutaric acid, $Na_2S_2O_4$, 3-hydroxybutyric acid, sodium maleate, malonic acid, methylene succinic acid, sodium pyrophosphate, succinic acid, succinimide and L-tartaric acid (Lot 071050) were used as supplied by British Drug Houses, Ltd., crystalline bovine serum albumin, NAD^+ , NADH, oxaloacetic acid, phenazine methosulfate were used as supplied by Sigma Chemical Co., L-cysteic acid was used as supplied by Light and Co., Ltd., L-glutamic acid was used as supplied by Hoffman-La Roche; α -oxoglutaric acid, and sodium pyruvate (Lot 650215) were used as supplied by Boehringer und Söhne. Potassium acetoacetate was kindly supplied by Dr. W. C. HÜLSMANN. Preparations of L-malic acid supplied by Fluka (Lot 58609 73K) and Nutritional Biochemical Corp. were found to contain 0.2–0.5% succinate, as measured by the O_2 uptake in a system containing 0.1 M phosphate buffer (pH 7.6), 2 mM Amytal, 3 mM arsenite, 1 mM EDTA, 94 mM malate and an active heart-muscle preparation. Similarly, these preparations reduced cytochrome *b* in these preparations in the presence of 10 mM cyanide. Succinic acid could also be detected in these preparations by paper (Whatman No. 1) chromatography, using chloroform–95% ethanol–conc. formic acid (100:100:24, v/v/v) as ascending solvent. After 36 h at 20°, the paper was dried at room temperature for 24 h and sprayed with ethanolic bromophenol blue¹⁷. The sample from Fluka was also found to contain fumaric acid by this technique. A sample of DL-malic acid (British Drug Houses, Ltd.) and of D-malic acid (Sigma Chemical Co., Lots M 32 B-214 and M 111 B-205) were found to contain traces of fumaric acid by this technique. The L-malic acid used as a substrate was purified by large-scale paper chromatography, and the concentration of the eluted acid was determined by titration. Rechromatography showed no detectable impurities.

Fumaric acid (British Drug Houses, Ltd.) was crystallized 3 times from ethanol before use.

The water used in preparing solutions was twice distilled in glass.

Absorption spectra

The spectra were recorded at a scanning speed of 5 m μ /sec (from 750 m μ to 320 m μ) in 1-cm cells at 25° in a Cary Model 14 recording spectrophotometer. Anaerobic spectra were measured in Thunberg spectrophotometric cuvettes, in which the contents were evacuated and refilled with O₂-free (by passing through alkaline pyrogallol) and dried (over silica gel) nitrogen, the whole process being repeated 5 times, ending with nitrogen. The spectra reported in this paper have been corrected for shifts in the base line during the recording, for dilution after additions and for absorption of the solutions added.

The enzyme was dissolved in 0.1 M potassium phosphate buffer (pH 7.6), containing 1 mM EDTA, for all spectrophotometric measurements.

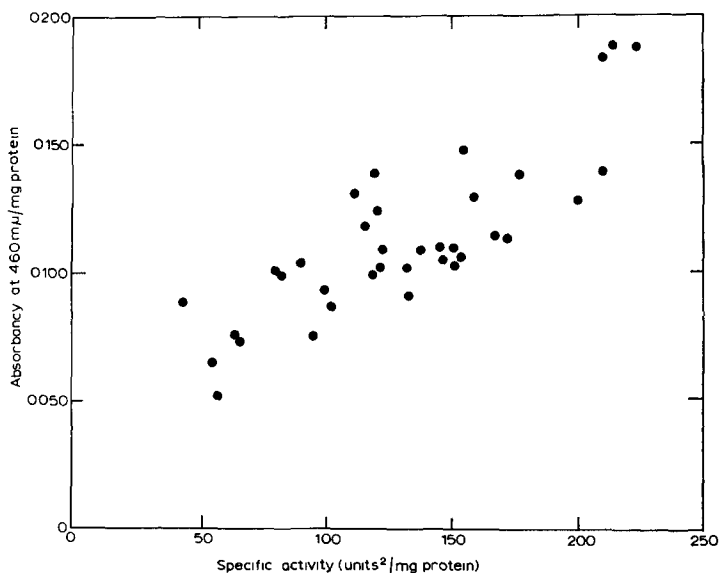


Fig. 1. The relationship of absorbance at 460 m μ expressed per mg protein to catalytic activity (determined by the ferricyanide method) in units²/mg protein.

RESULTS

Absorption spectrum of isolated succinate dehydrogenase

Fig. 1 shows that there is a reasonable correlation between the absorbance at 460 m μ of different preparations and the catalytic activity. These measurements were made either directly after isolation of the preparation or after a short storage in liquid nitrogen, where the enzyme is stable for 3 weeks.

On standing at 25°, there is a rapid decline in absorbancy over the whole range measured (Fig. 2). This loss of absorbancy is about 50% more rapid under aerobic conditions (Fig. 2A) than under anaerobic (Fig. 2B). The enzyme also rapidly loses activity under aerobic conditions, the decline amounting to 40% in 44 min under the conditions of the experiment shown in Fig. 2A. Despite the absorbancy changes also observed under anaerobic conditions, the activity was found to increase by 22% after standing for 3 h at 25°.

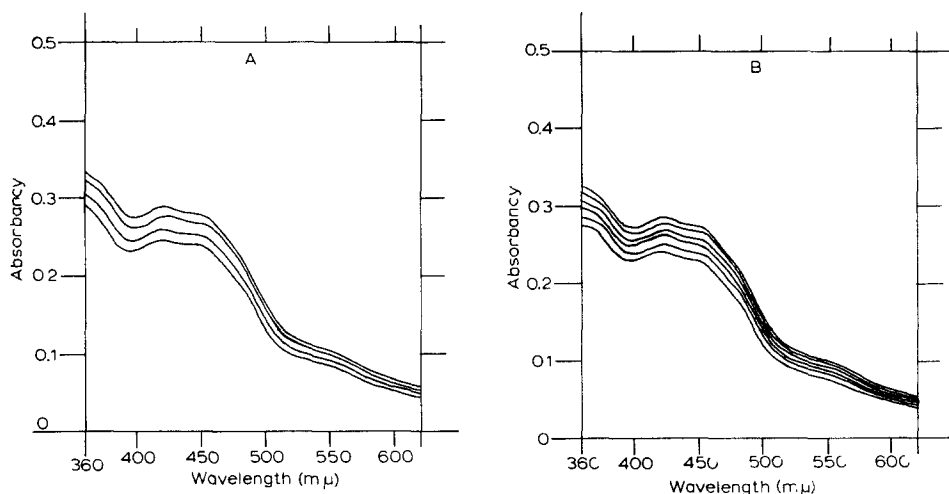


Fig. 2. The loss of absorbancy of succinate dehydrogenase on standing at 25°. (A) Enzyme (140 units²/mg protein, 1.90 mg protein/ml) was kept at 25° in the presence of air and the spectra recorded at zero time and after 10, 28 and 44 min. The four spectra are shown from top (zero time) to bottom (44 min). The specific activity of the enzyme was 119, 109 and 85 after 10, 28 and 44 min. (B) Enzyme (140 units²/mg protein, 1.85 mg protein/ml) was kept at 25° under anaerobic conditions and the spectra recorded at zero time and after 10, 30, 45, 105 and 164 min. The six spectra are shown from top (zero time) to bottom (164 min). After 192 min the specific activity was 171 units²/mg protein.

Effect of competitive inhibitors on the spectrum of succinate dehydrogenase

Fig. 3 (A and B) shows the effect of succinate and fumarate, added separately and together, on the absorption spectrum of the enzyme. Succinate caused a decline in the spectrum over the whole range between 400 mμ and 600 mμ, the maximum decline being at 460 mμ. Fumarate also caused a decline at this wavelength (maximum decline at 450 mμ), but the decline was much less than that obtained with succinate, while there was an increase at higher wavelengths, the maximum in the difference spectrum being found at 510 mμ, where there was an increase of 18.5% in the absorbancy. The spectrum with succinate + fumarate lies between those with succinate and fumarate alone.

In agreement with previous workers^{2,6}, the addition of Na₂S₂O₄ caused a much greater decline of absorbancy than succinate (Fig. 4). Also in agreement with MASSEY AND SINGER⁶, the addition of fumarate to the Na₂S₂O₄-reduced enzyme caused a substantial, but incomplete, recovery of the absorption at 460 mμ. The difference

spectrum in Fig. 4B, which clearly shows the peak at $510\text{ m}\mu$, gives a partial explanation of this incomplete recovery.

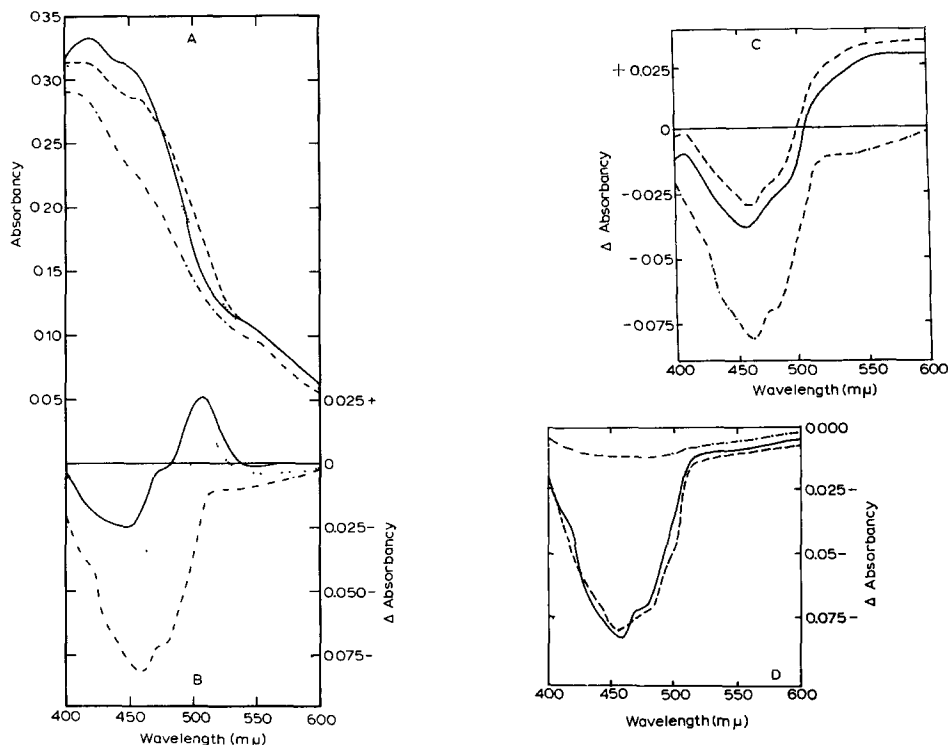


Fig. 3. The effect of succinate and of competitive inhibitors on the absorption spectra of succinate dehydrogenase. A preparation containing 192 units²/mg protein and 2.3 mg protein/ml was used. (A) —, no addition; ---, 10 mM succinate; ----, 100 mM fumarate; ·····, 10 mM succinate added after 100 mM fumarate. (B) Difference spectra calculated from A. —, treated with succinate minus untreated; ---, treated with fumarate minus untreated; ·····, treated with fumarate then succinate minus untreated. (C) —, treated with succinate minus untreated; ----, treated with oxaloacetate (100 mM) minus untreated; —·—·—, treated with oxaloacetate then succinate minus untreated. (D) —, treated with succinate minus untreated; ---, treated with α -oxoglutarate (100 mM) minus untreated; - - - - -, treated with α -oxoglutarate then succinate minus untreated.

Malonate, methylene succinate, maleate and acetoacetate had similar effects on the absorption spectrum of the enzyme (Fig. 5). These compounds are all competitive inhibitors of succinate dehydrogenase and contain either an activated methylene group or a system of π electrons such as that associated with a $C=C$ bond. They are, therefore, all electron donors.

Oxaloacetate, the most powerful inhibitor of succinate dehydrogenase¹⁸, had a different effect on the spectrum (Fig. 5). There was a much larger decrease of absorption in the blue with a maximum decrease at $460\text{ m}\mu$ and a minimum decrease at $410\text{ m}\mu$. There was also a much larger increase in the longer-wavelength region than obtained with the other competitive inhibitors. The maximum increase was also at

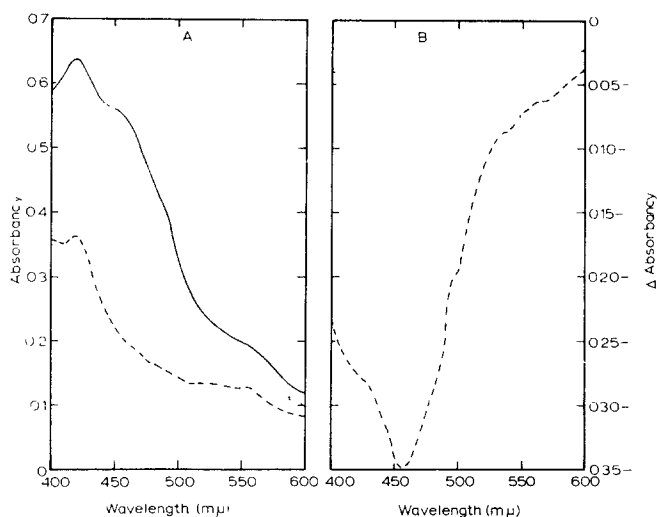


Fig. 4. The effect of fumarate on $\text{Na}_2\text{S}_2\text{O}_4$ -reduced enzyme. Succinate dehydrogenase containing 1.45 units/mg protein and 5.5 mg protein/ml was used. (A) —, untreated enzyme; - - - - -, +0.84 mM $\text{Na}_2\text{S}_2\text{O}_4$, $\text{Na}_2\text{S}_2\text{O}_4$ followed by 120 mM fumarate (B) Difference spectra calculated from A - - - - -, treated with $\text{Na}_2\text{S}_2\text{O}_4$ minus untreated; , treated with $\text{Na}_2\text{S}_2\text{O}_4$ followed by fumarate minus untreated

a higher wavelength (580 $\text{m}\mu$), the increase amounting to 52% of the absorption of the untreated enzyme. Increased absorption was found up to 750 $\text{m}\mu$. D- and L-malate had the same effect as oxaloacetate, except that the spectral change at the higher wavelengths occurred more slowly (instantaneous with oxaloacetate, 3–5 min with L-malate, 10–15 min with D-malate). It will be shown in a later paper¹⁹ that D- and L-malate are slowly converted to oxaloacetate by the enzyme and that the decrease of absorption in the region of 460 $\text{m}\mu$ represents an actual hydrogen or electron transfer

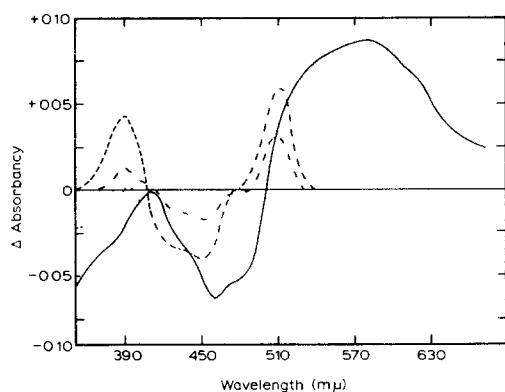


Fig. 5. The effect of competitive inhibitors on the absorption spectra of succinate dehydrogenase. Enzyme containing 233 units/mg protein and 6.5 mg protein/ml was used. . . . , treated with 70 mM oxaloacetate minus untreated, - - - - -, treated with 100 mM acetoacetate, malonate, maleate, methylene succinate or fumarate minus untreated, measured between 410 $\text{m}\mu$ and 700 $\text{m}\mu$; , 100 mM malonate measured minus untreated between 350 $\text{m}\mu$ and 410 $\text{m}\mu$. - - - - -, maleate, methylene succinate or fumarate minus untreated, measured between 350 and 410 $\text{m}\mu$.

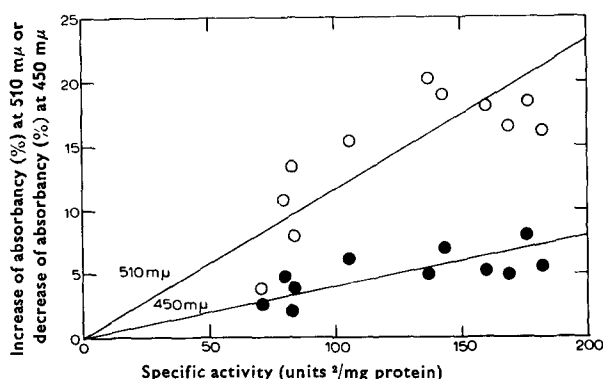


Fig. 6. The relationship between specific activity (units²/mg protein) and the increase of absorption (%) at 510 mμ or decrease of absorption (%) at 450 mμ upon addition of 100 mM fumarate to succinate dehydrogenase.

to the flavin, whereas the slower increase of absorption at higher wavelengths is due to the formation of a complex of the enzyme with the oxaloacetate formed.

Other compounds structurally related to succinate, but which are very weak competitive inhibitors (if at all) were found to have little effect on the spectrum. The compounds tested with the K_i , determined as described in METHODS, in brackets were: alloxan (30 mM), L-aspartate (280 mM), citrate (40 mM), crotonate (∞), L-cysteate (600 mM), diacetyl (380 mM), glycerol (∞), L-glutamate (350 mM), glutarate (40 mM), 3-hydroxybutyrate (∞), α -oxobutyrate (150 mM), α -oxoglutarate (70 mM), pyruvate (130 mM), succinimide (260 mM), D-tartrate (200 mM), L-tartrate (170 mM), *meso*-tartrate (200 mM)*.

Pyrophosphate, which is also a powerful competitive inhibitor²⁰, but which is

TABLE I
COMPARISON OF INHIBITOR CONSTANTS (K_i) DETERMINED KINETICALLY
AND DISSOCIATION CONSTANTS (K_D) DETERMINED SPECTROPHOTOMETRICALLY

Inhibitor	K_D (mM)	K_i (mM)
Malonate	0.028	0.018*
Fumarate	2.6	1.3*
Methylene succinate	6.3	1.8
Maleate	7.0	6.2
Acetoacetate	80	40
Oxaloacetate	0.0040	0.0015*

* These values for which we thank Mr. W. ZEYLEMAKER were determined with the soluble enzyme as described in METHODS.

* Some of these compounds were found to have small but reproducible effects on the absorption spectrum. Alloxan caused a slight decrease (about 3%) in the absorption over the whole range 400–700 mμ; L-aspartate decreased the absorption in the region 470–500 mμ (by 2.6% at 495 mμ); crotonate decreased the absorption at 490–500 mμ by 2.5%; diacetyl decreased the absorption at 395–410 mμ (by 6% at 405 mμ); glutarate decreased the absorption in the region 455–510 mμ (by 6% at 500 mμ); 3-hydroxybutyrate increased the absorption at 490–520 mμ by 2–4% in some preparations; α -oxoglutarate decreased the absorption at 460 mμ by 4%; succinimide decreased the absorption in the region 430–510 mμ (by 3–4% at 470 mμ).

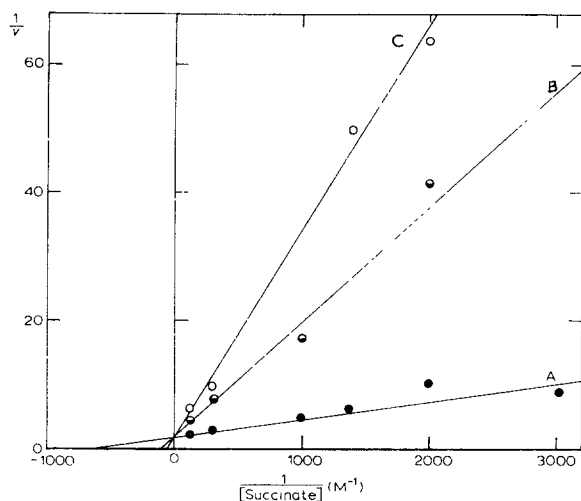


Fig. 7. Determination of the inhibitor constant (K_i) for maleate by the reciprocal-plot method. A, B, C represent absence of maleate, 35 and 70 mM maleate respectively

not in any obvious way structurally related to succinate, also had no effect on the spectrum.

The decrease of absorption at 450 m μ and the increase at 510 m μ , obtained on adding fumarate (Fig. 6) to different preparations was roughly proportional to the specific activity of the preparation. Similar results were obtained at 350–420 m μ and with malonate, maleate, methylene succinate and acetoacetate.

In Table I the kinetically determined inhibition constants (K_i) are compared

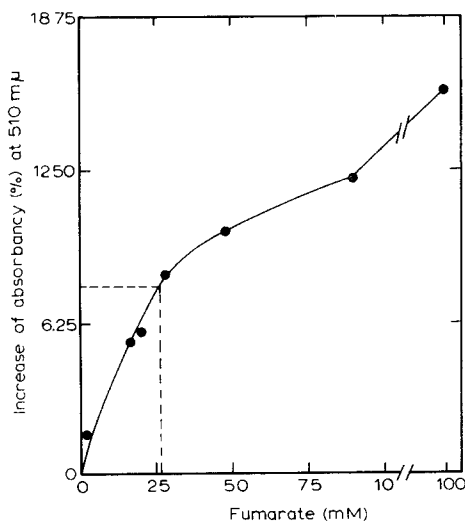


Fig. 8. Determination of the dissociation constant (K_D) by spectral titration. The enzyme used in these determinations contained 5.5 mg protein/ml and had a specific activity of 197 units²/mg protein.

with the spectrophotometrically determined dissociation constants (K_D). The determination of K_I is illustrated for maleate in Fig. 7, and of K_D is illustrated for fumarate in Fig. 8. In disagreement with HELLERMAN *et al.*²¹, but in agreement with FRANKE AND HOLZ²², maleate was found to be a competitive inhibitor. The effectiveness of methylene succinate as a competitive inhibitor was also first reported by the latter workers.

Fig. 3 (A and B) illustrates that when succinate is added to the fumarate-enzyme complex, there is a decrease of absorbancy characteristic of the changes which occur when succinate is added to the untreated enzyme. The magnitude of these changes is, however, less than that obtained with succinate alone, and the spectrum shows that some fumarate complex is still present. On the other hand, with oxaloacetate. There was only a small change on adding succinate, as is to be expected since the dissociation constant of the oxaloacetate complex is less than that of the fumarate complex (Fig. 3C). When succinate was added after the much less firmly bound α -oxoglutarate, the spectrum obtained was the same as with succinate alone (Fig. 3D).

DISCUSSION

In the absence of an electron or hydrogen acceptor, the reactions of succinate dehydrogenase (E) with succinate (S) and with the competitive inhibitors (I) can be formally represented by the equations



where nothing is implied about the chemical nature of the compounds.

In the absence of competitive inhibitors, the characteristic spectral change with substrate (see Fig. 3) corresponds with the reduction of a flavin molecule. This is supported by electron-spin resonance studies²³⁻²⁵ which will be reported in detail in another paper²⁶.

Characteristic spectral changes, differing from those obtained with succinate, have been demonstrated on the addition of a number of competitive inhibitors to the enzyme. The general agreement between the K_I , the inhibition constant determined by kinetic experiments with the particulate or soluble succinate dehydrogenase, and the K_D , the dissociation constant of the complex determined as the amount of inhibitor necessary for half-maximal spectral response, is strong support for the view that the spectra obtained are those of the EI complexes. The fact that the subsequent addition of succinate to the fumarate-treated enzyme has a spectral response similar in character but of smaller magnitude to that obtained with succinate alone shows that Eqn. 2 is reversible.

A spectral change similar to those obtained with the competitive inhibitors (except oxaloacetate) was previously reported by KEARNEY^{27,28} with malonate, and ascribed by her to a change of the protein occurring during the activation of the enzyme by malonate*. She had shown that succinate dehydrogenase isolated by the

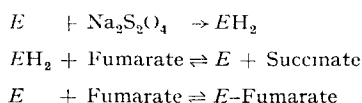
* There are, however, important differences in detail between the spectral changes obtained on adding malonate to the enzyme (Fig. 5) and the changes reported by KEARNEY. The increase

method of SINGER *et al.*¹ was not fully active unless preincubated with succinate, fumarate, malonate or phosphate. It is not likely that activation is an important factor in our work, since the enzyme prepared by the procedure of WANG *et al.*² involving pre-treatment with succinate, is already almost maximally activated. Moreover, the spectral shifts observed were complete within 45 sec, in contrast to the 30 min required for activation of the SINGER enzyme^{27,28}, and are immediately reversed on the addition of succinate.

KIMURA *et al.*²⁹ have recently discussed the spectral phenomena again in terms of activation as the main factor although they agree that another process may be involved. Fig. 6 shows that the height of the 510-m μ band is dependent on the specific activity of the enzyme. Our explanation differs fundamentally from that of KIMURA *et al.*²⁹ since we believe that the inhibitor complex is formed with an active enzyme. Activation, which makes the enzyme more suitable for forming the complex, is of secondary importance.

The characteristic spectral effect of oxaloacetate, which differs markedly from that of the other competitive inhibitors, can be used as a sensitive qualitative test for oxaloacetate. By using this test, we were able to show for the first time that L-malate and D-malate are oxidized by the enzyme¹⁹. We could also show the presence of fumarate hydratase (EC 4.2.1.2) in the purified succinate dehydrogenase preparation by adding fumarate and NAD⁺ (the purified enzyme usually contains a trace of malate dehydrogenase, EC 1.1.1.37). The oxaloacetate-enzyme complex was formed. The characteristic peak at 390 m μ in the difference spectrum of the fumarate complex (Fig. 5) was also used to reveal the presence of a trace of fumarate in the preparation of D-malate used.

The formation of the enzyme-fumarate complex, which absorbs at 460 m μ less than that of the untreated enzyme, provides the explanation of the fact, discussed in the INTRODUCTION, that the addition of fumarate to the Na₂S₂O₄-reduced enzyme does not completely restore the absorbancy at 460 m μ to that of the untreated enzyme^{6,7}. The reactions which take place are probably



It is interesting that pyrophosphate, which is also a competitive inhibitor, does not show the spectral response of these competitive inhibitors which are structurally related to succinate. It seems likely that the latter type of competitive inhibitors reacts with the same site on the enzyme as succinate itself. However, any compound which reacts with the free enzyme but not with the *ES* compound behaves kinetically as a competitive inhibitor. It is possible that pyrophosphate reacts with a component (maybe an iron atom) which is in a reactive form in the free (oxidized) enzyme but not in the *ES* compound (reduced enzyme).

The chemical nature of the *EI* complexes is not known. As already mentioned, all competitive inhibitors which affect the spectrum of succinate dehydrogenase have

of absorbancy between 350 and 400 m μ , the double maximum in the region 470-550 m μ and the decrease of absorption beyond 540 m μ , reported by KEARNEY, are not seen in the difference spectrum shown in Fig. 5. The maximum increase in absorbancy (at 510 m μ) is also considerably greater in our experiments

electron-donating properties. It is possible that the bands formed are due to charge-transfer complexes between these electron-donating groups and the flavin. The diffuse difference band obtained with oxaloacetate is certainly consistent with this possibility.

In connection with the fact that only strongly bound electron donors have these spectral effects, it is of interest to note that MULLIKEN³⁰ has pointed out that reactants which are normally capable of forming weak charge-transfer complexes show a large increase in such complex formation during compression, *e.g.* by dipole forces.

It seems probable that the inhibitors fit closely in the active site forming perhaps an inclusion compound. In this inclusion compound the bound inhibitor can give rise to changes in polarity near the flavin, the degree of change being dependent upon the nature of the inhibitor, particularly on the structure of the aliphatic linkage between the carboxylic acid groups. The work of HARBURY *et al.*³¹ has shown that changes in polarity of the solvent cause marked changes in the absorption spectrum of flavins. A decrease of polarity brings about (a) a shift of the 370-m μ band to lower wavelengths with a deepening of the minimum at about 390 m μ , (b) a shift of the 450-m μ band to higher wavelengths with the formation of shoulders on both sides of the band, most markedly to the red. These changes lead to a negative difference spectrum (apolar minus polar) in the near ultraviolet and at about 450 m μ with a positive difference at 480–500 m μ . The changes in the spectra especially in the near ultraviolet obtained on adding malonate or oxaloacetate to succinate dehydrogenase resemble closely what would be expected from polarity changes in the region of the flavin prosthetic group. Upon malonate addition, the difference spectrum shows a minimum decrease at 390 m μ . Addition of oxaloacetate, which compared to malonate shows an increase in polarity, causes a 20-m μ bathochromic shift in the difference spectrum with a minimum decrease at 410 m μ . Fumarate, maleate, methylene succinate and acetoacetate differ in that there is positive absorption at about 390 m μ in the difference spectrum, although the changes between 410 m μ and 520 m μ are exactly the same as for malonate. HARBURY^{31,33} and co-workers have pointed out that the spectral changes in the 330–400-m μ and in the 450–500-m μ region resulting from changes in polarity are not directly related to one another. Further it may be noted that polarity of the solvent has a marked influence on the position of the charge-transfer absorption band^{31,32}. It is conceivable that the 390-m μ band obtained with fumarate, etc. is caused by a dipole induced in the C=C double bond, which is not present in malonate, but on the other hand, the latter has a permanent dipole round the central carbon atom. It certainly cannot be ruled out that a combination of charge-transfer complex formation and polarity changes is responsible for the phenomena discussed here. The spectral effects with the non-specific compounds tested indicate that other interactions involving protein and/or flavin may be involved.

KOSTER AND VEEGER³⁴ have found that the spectral changes associated with the forming of a complex between D-amino acid oxidase (EC 1.4.3.3) and its competitive inhibitor, benzoate (*cf.* YAGI³⁵), resemble those obtained on adding competitive inhibitors to succinate dehydrogenase. With L-amino acid oxidase (EC 1.4.3.2) DE KOK AND VEEGER³⁶ have found a relationship between the spectrally determined dissociation constant and the kinetically determined inhibitor constant with *o*-amino-benzoate. These spectral changes can also be interpreted in terms of the formation of a charge-transfer complex or in terms of changes in polarity in the region of the flavin. VEEGER AND MASSEY³⁷ have also interpreted the shift of the peak from 455 m μ

to 452 $m\mu$, and the weakening of the shoulder at 480 $m\mu$, which occurs when liponamide dehydrogenase is incubated with Cu(II), in terms of changes of polarity round the flavin, possibly as a result of oxidation of -SH groups.

The decline of absorbancy of the enzyme over the whole spectral range studied which occurs on standing at 25° is a curious phenomenon for which no explanation is available. The relative decline in absorbancy is almost independent of the wavelength.

The addition of succinate to our enzyme preparation brought about a considerably larger decrease in the absorbancy between 400 and 500 $m\mu$ (27.3% at 460 $m\mu$) than that reported by SINGER *et al.*¹ and WANG *et al.*². The decrease was only slightly greater than that reported recently by KING³⁸. The absorbancy declines only slightly between 500 and 600 $m\mu$. Na₂S₂O₄ causes a considerably greater decrease over the whole range 400–600 $m\mu$. Electron-spin resonance studies^{23–25} show that Na₂S₂O₄ causes complete reduction of the flavin to the leuco form whereas succinate reduces it only to the stage of the semiquinone. Increased absorption in the 500–600- $m\mu$ region, characteristic for the formation of a semiquinone^{9,39–44}, is not observed on the addition of substrate. As BEINERT⁴⁵ has noted, this is also the case with other metalloflavoproteins. However, any increase would be masked by the general decrease in absorbancy which occurs when either succinate or Na₂S₂O₄ is added to the enzyme. Indeed, the relatively slight decline of absorbancy between 500 and 600 $m\mu$ found with succinate compared with Na₂S₂O₄ suggests quite strongly that a chromophore is formed in this region with succinate. It is probable that the iron, possibly chelated with the flavin (*cf.* refs. 40–42), contributes considerably to the spectrum of the enzyme.

Evidence will be presented in a later paper that some of the absorption in the 500–600- $m\mu$ region might be due to sulphur combined to the iron atom which is also chelated to the flavin. This is of interest in relation to the presence of acid-labile sulphur in succinate dehydrogenase⁴⁶ and other metalloflavoproteins^{47,48}, as well as the interesting metal aromatic dithiol complexes of GRAY AND BILLIG⁴⁹ showing asymmetric electron-spin resonance signals in the solid state. Also AASA *et al.*⁵⁰ have shown that binding of Fe(III) or Cu(II) to conalbumin leads to considerable increase in absorption in the visible part of the spectrum (*cf.* ref. 51).

Fig. 4 shows that the enzyme, either untreated or reduced with Na₂S₂O₄ or succinate, has an absorption band at about 555 $m\mu$. This is most noticeable in the enzyme reduced with Na₂S₂O₄. The grouping responsible for this band is unknown.

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REFERENCES

- ¹ T. P. SINGER, E. B. KEARNEY AND P. BERNATH, *J. Biol. Chem.*, **223** (1956) 599.
- ² T. Y. WANG, C. L. TSOU AND Y. L. WANG, *Sci. Sinica Peking*, **5** (1956) 73.
- ³ T. Y. WANG, C. L. TSOU AND Y. L. WANG, *Sci. Sinica Peking*, **7** (1958) 65.
- ⁴ E. B. KEARNEY, *J. Biol. Chem.*, **235** (1960) 865.
- ⁵ V. MASSEY AND C. VEEGER, *Biochim. Biophys. Acta*, **48** (1961) 33.
- ⁶ V. MASSEY AND T. P. SINGER, *J. Biol. Chem.*, **228** (1957) 263.
- ⁷ D. KEILIN AND T. E. KING, *Proc. Roy. Soc. London Ser. B.*, **152** (1960) 163.
- ⁸ D. V. DERVARTANIAN AND C. VEEGER, *Biochem. J.*, **84** (1962) 65P.
- ⁹ C. VEEGER, *14. Mosbach-Colloquium der Deutschen Physiologischen Gesellschaft, 1963, Structure and Activity of Flavoproteins of Respiratory Chain*, Springer Verlag, Heidelberg, 1964, p. 157.
- ¹⁰ D. KEILIN AND E. F. HARTREE, *Biochem. J.*, **41** (1947) 503.
- ¹¹ E. C. SLATER, *Biochem. J.*, **45** (1949) 1.
- ¹² D. KEILIN AND E. F. HARTREE, *Proc. Roy. Soc. London Ser. B.*, **124** (1938) 397.
- ¹³ A. G. GORNALL, C. J. BARDRAWILL AND M. M. DAVID, *J. Biol. Chem.*, **177** (1949) 751.
- ¹⁴ E. B. KEARNEY AND T. P. SINGER, *J. Biol. Chem.*, **219** (1956) 963.
- ¹⁵ T. E. KING, *J. Biol. Chem.*, **238** (1963) 4032.
- ¹⁶ E. C. SLATER AND W. B. BONNER, *Biochem. J.*, **52** (1952) 185.
- ¹⁷ J. B. STARK, J. B. GOODBAN AND H. S. OWENS, *Anal. Chem.*, **23** (1951) 413.
- ¹⁸ N. B. DAS, *Biochem. J.*, **31** (1937) 1124.
- ¹⁹ D. V. DERVARTANIAN AND C. VEEGER, to be published.
- ²⁰ C. L. TSOU, *Biochem. J.*, **49** (1951) 512.
- ²¹ L. HELLERMAN, O. K. REISS, S. S. PARMAR, J. WEIN AND N. L. LASSER, *J. Biol. Chem.*, **235** (1960) 2468.
- ²² W. FRANKE AND E. HOLZ, *Ann. Chem.*, **608** (1957) 168.
- ²³ H. BEINERT AND R. H. SANDS, *Biochem. Biophys. Res. Commun.*, **3** (1960) 41.
- ²⁴ T. E. KING, R. L. HOWARD AND H. S. MASON, *Biochem. Biophys. Res. Commun.*, **5** (1961) 329.
- ²⁵ D. V. DERVARTANIAN, C. VEEGER AND J. D. W. VAN VOORST, *Biochim. Biophys. Acta*, **73** (1963) 660.
- ²⁶ J. D. W. VAN VOORST, C. VEEGER AND D. V. DERVARTANIAN, to be published.
- ²⁷ E. B. KEARNEY, *J. Biol. Chem.*, **229** (1957) 363.
- ²⁸ E. B. KEARNEY AND T. P. SINGER, *Federation Proc.*, **16** (1957) 203.
- ²⁹ T. KIMURA, J. HAUBER AND T. P. SINGER, *Biochem. Biophys. Res. Commun.*, **11** (1963) 83.
- ³⁰ R. S. MULLIKEN, *J. Am. Chem. Soc.*, **74** (1952) 811.
- ³¹ H. A. HARBURY, K. F. LANOUE, P. A. LOACH AND R. M. AMICK, *Proc. Natl. Acad. Sci. U.S.*, **45** (1959) 1708.
- ³² E. M. KOSOWER, in P. D. BOYER, H. LARDY AND K. MYRBÄCK, *The Enzymes*, Vol. 3, Academic Press, London, 1963, 2nd ed., p. 171.
- ³³ H. A. HARBURY AND K. A. FOLEY, *Proc. Natl. Acad. Sci. U.S.*, **44** (1958) 662.
- ³⁴ J. F. KOSTER AND C. VEEGER, to be published.
- ³⁵ K. YAGI AND T. OZAWA, *Biochim. Biophys. Acta*, **67** (1963) 319.
- ³⁶ A. DE KOK AND C. VEEGER, to be published.
- ³⁷ C. VEEGER AND V. MASSEY, *Biochim. Biophys. Acta*, **64** (1962) 83.
- ³⁸ T. E. KING, *J. Biol. Chem.*, **238** (1963) 4037.
- ³⁹ H. BEINERT, in P. D. BOYER, H. LARDY AND K. MYRBÄCK, *The Enzymes*, Vol. 2, Academic Press, New York, 1960, 2nd ed., p. 339.
- ⁴⁰ P. HEMMERICH, *Helv. Chim. Acta*, **47** (1964) 464.
- ⁴¹ P. HEMMERICH, *14. Mosbach-Colloquium der Deutschen Physiologischen Gesellschaft, 1963, Wirkungsmechanismen von Enzymen*, Springer Verlag, Heidelberg, 1964, p. 183.
- ⁴² P. HEMMERICH, *Habilitationsschrift*, Basel, 1963.
- ⁴³ V. MASSEY AND Q. H. GIBSON, *Federation Proc.*, **23** (1964) 18.
- ⁴⁴ V. MASSEY AND C. VEEGER, *Ann. Rev. Biochem.*, **32** (1963) 579.
- ⁴⁵ H. BEINERT, *Proc. Intern. Congr. Abstr.*, New York, 1964, Vol. IV, 1964, p. 285.
- ⁴⁶ V. MASSEY, *Biochim. Biophys. Acta*, **30** (1958) 500.
- ⁴⁷ P. HANDLER, K. V. RAJAGOPALAN AND V. ALEMAN, *Federation Proc.*, **23** (1964) 30.
- ⁴⁸ K. V. RAJAGOPALAN AND P. HANDLER, *Federation Proc.*, **23** (1964) 162.
- ⁴⁹ H. B. GRAY AND E. BILLIG, *J. Am. Chem. Soc.*, **85** (1963) 2019.
- ⁵⁰ R. AASA, B. G. MALMSTRÖM, P. SALTMAN AND T. VÄNNGÅRD, *Biochim. Biophys. Acta*, **75** (1963) 203.
- ⁵¹ J. S. RIESKE, D. H. MACLENNAN AND R. COLEMAN, *Biochem. Biophys. Res. Commun.*, **15** (1964) 338.